ASSIGNMENT 3

Q6 - Pyramid Pattern Generator

JavaScript solution

```
// q6.js
"use strict";

const input = 5; // Replace with Number(prompt("Enter row count:")) in a browser.
const rows = Number.isInteger(input) && input > 0 ? input : 5;

for (let row = 1; row <= rows; row++) {
  let line = "";
  for (let col = 1; col <= row; col++) {
    line += "* ";
  }
  console.log(line.trimEnd());
}

// var comparison:
// for (var i = 1; i <= 4; i++) {
//   var line2 = "";
//   for (var j = 1; j <= i; j++) line2 += "* ";
//   console.log(line2.trimEnd());
// }
// var is function-scoped, so i, j, and line2 are not block-scoped.
// let is block-scoped. In this synchronous loop, both versions print
// the same pattern, but scope/reuse can matter after the loop.
// Strict mode catches assignments to undeclared variables.
```

Explanation / Debug notes

The outer loop controls the number of rows; the inner loop adds one star per column. The default row count is 5. In a browser, prompt() can collect input; in Node.js, use a command-line input method or change input directly. Replacing let with var usually keeps this output the same, but var is function-scoped rather than block-scoped.

Run the code in the browser console or Node.js. Save each solution in its own .js file as required.